

Fig. 3: Connection of the DC to DC converter to op-amp

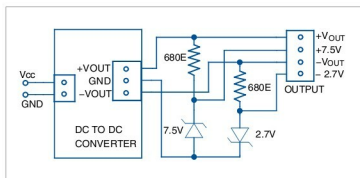


Fig. 4: Connection of the DC to DC converter to Zener diode

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- D1-D4 - 1N4148 signal diode
- Resistors (all 1/4-watt, $\pm 5\%$ carbon):
- R1, R2 - 10-kilo-ohm
- R3 - 1-ohm

Capacitors:

- C1 - 3.3nF ceramic disk
- C2, C6, C7 - 0.1uF ceramic disk
- C3-C5, C8, C10 - 47uF, 25V electrolytic
- C9 - 0.01uF ceramic disk

Miscellaneous:

- CON1 - 6-pin connector
- CON2 - 2-pin connector
- 9-15V DC adaptor

(in Fig. 1) are important to check output current of the timer at CON2. This current should always be below the maximum, which is 200mA for NE555.

Fig. 2 shows how to connect the DC to DC converter to a wall adaptor, MCU kit and input/output (I/O) expansion board.

Fig. 3 shows how to connect

popular operational amplifiers to the DC to DC converter board. Output voltages of the converter depend on the load. That is not a problem in many cases and, if you want, you can restrict and

regulate these to particular pre-defined levels.

Fig. 4 shows how to regulate output voltages of the DC to DC converter with low-cost Zener diodes like 1N4737 (7.5V), BZX-85C2V7 (2.7V), BZX55C2V4 (2.4V) and so on. This circuit shows two different Zener diodes to get regulated voltages such as +7.5V and -2.7V.

Construction and testing

An actual-size, single-side PCB for the DC to DC converter circuit (Fig. 1) is shown in Fig. 5 and its component layout in Fig. 6. Enclose the PCB in a suitable box and provide connectors for 230V AC adaptor for input and CON1 for output.

It is preferable to select the working frequency above 20kHz. (During testing we have used around 7kHz frequency. But that depends on the requirements for

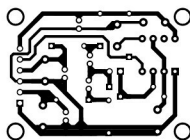


Fig. 5: Actual-size PCB pattern of the DC to DC converter

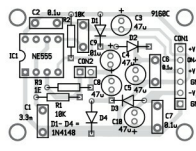


Fig. 6: Component layout of the PCB

the power supply of the embedded system and parameters of the components in the converter.) The circuit does not need any adjustment and will start working immediately after proper connections are done.

It is recommended to use fast-switching, low-voltage-drop diodes. 1N4148 would work but 1N5819 or similar is preferred. 1N400X series diodes are too slow for most applications but will work at low switching frequencies. Capacitors should be used for high-frequency switching power supplies.

Note. PCB layout for the circuit in Fig. 1 only is shown here. Remaining circuits (Figs 2, 3 and 4) can be wired on general-purpose PCBs.

Parts list is also of the circuit shown in Fig. 1 only. **EFY**



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